

UNIT II

OPTICS

Interference - Air Wedge - Mechelson's interferometer - Wave length Determination - Interference Filter - Antireflection Coatings. Diffraction - Diffraction grating - Dispersive power of grating - Resolving Power of Grating and Prism Polarisation - Huygens Theory of Double Refraction - Quarter and Half Wave Plates - Specific Rotary Power Laurent Half Shade Polarimeter.

Pondicherry University 2 marks Question& Answer

1. What is an air wedge? (Jan 2010, April 2010)

A wedge shaped air film is enclosed in between two glass plates is known as airwedge.

2. What is quarter wave plate? (Jan 2014, May 2017)

Quarter wave plate is doubly refracting material made of uniaxial crystal. Whose refracting faces are cut parallel to the optic axis. The thickness of this plate is in such a way that it produces a path difference of $\lambda/4$ or a phase difference of $\pi/2$ between the ordinary and extra ordinary rays emerging from it.

3. What is an anti reflection coating? (Jan 2011, 2016, May 2012, May 2019, Dec 2019)

A transparent material whose refractive index value lies between air and glass is coated on the surface the glass in order to prevent or to reduce reflection is called as anti reflection coating.

4. Explain the term Resolving power of an optical instrument. (Jan 2011, May 2019, Dec 2019)

The ability of the instrument to show the separation between the spectral lines of two wavelengths, which are very near to each other. $R. P = \frac{\lambda}{d\lambda}$

5. What is a circularly polarized light? (Jan 2010)

If light is composed of two plane waves of equal amplitude by differing in phase by 90° , then the light is said to be circularly polarized

6. What are uniaxial crystals? (May 2011)

A Uniaxial crystal is the one which has only one optic axis. A Bi-axial crystal is the crystal which has two optic axis.

7. What is half wave plate? (April, 2010, May 2011, May 2018, Sep 2020)

Half wave plate is a doubly refracting material made of uniaxial crystal whose refracting faces are cut parallel to the optics axis. The thickness of this plate is in such a way that it produces a path difference of $\lambda/2$ or a phase difference of π between ordinary ray and extraordinary ray emerging from it. The thickness of the half wave plate is to produce an optical path difference of $\lambda/2$ and is given by $t = \lambda / 2 (\mu_o - \mu_e)$

8. Explain the term specific rotation? (Jan 2013, Dec 2018, May 2019)

The specific rotation is defined as the rotation produced by a decimetre (10cm) long column of the liquid containing 1 gram of the active substance in one cc of the solution. It is given by $S = 10\theta / LC$ Where S – Specific rotation θ – angle of rotation L – length of the solution in cm and C is the concentration of the solution,

9. State the principle of Superposition of light.

(May 2012, April/May 2011, May 2017)

The principle states that resultant displacement of a particle of the medium acted upon by two or more waves simultaneously is the algebraic sum of the displacements of the same particle due to individual waves, in the absence of the others.

10. Define resolving power of a prism? (May 2013)

The resolving power of a prism is defined as its capacity to form separate spectral lines of two wavelengths which are very near to each other. It is given by

$$\frac{\lambda}{d\lambda} = t \cdot \frac{d\mu}{d\lambda}$$

11. What do you understand by double refraction? (May 2013, Jan 2014)

When a light is passed through a calcite crystal, the single incident beam splits into two reflected rays (i) Ordinary ray (O-ray) (ii) Extra-ordinary ray (E-ray)

12. Distinguish between interference and diffraction? (Jan 2013, Jan 2014, Dec 2018, Dec 2019)

Sl.No	Interference	Diffraction
1	Interference takes place when two different coherent wave-fronts originating from the same source superimpose each other.	Diffraction pattern is the result of interaction of light coming from different parts of the same wave-front.
2	Interference bands are of equal width	Interference bands are not of equal width
3	All bright bands are of uniform intensity	All bright bands are not of the same intensity
4	Points of minimum intensities are perfectly dark	Points of minimum intensities are not perfectly dark

13. In an air wedge experiment, if the air gap between the wedge forming plates is filled with water what will happen to the fringe-width? (Jan 2014, May 2018)

The fringe-width decreases because the fringe width β is inversely proportional to the refractive index which is given as $\beta = \lambda / 2\mu\theta$.

14. Mention the applications of quarter wave plate (Jan 2016)

Applications :

(i) When the vibrations of the polarised light, inclined at an angle of 45° to the optics is made to incident on a quarter wave plates the emergent beam becomes circularly polarised light. Therefore quarter wave plate is used to convert plane polarised light to circularly polarised light and vice versa.

(ii) When the vibrations of the polarised light inclined at an angle not equal to 45° (i.e. other than 45°) to the optic axis is made to incident on a quarter wave plate, the emergent beam becomes elliptically polarised light. Thus it is used to convert plane polarised light to elliptically polarised light and vice versa.

Thus a quarter wave plate can be used to convert.

(i) Plane polarised light $\rightleftharpoons$ Elliptically polarised light

(i) Plane polarised light $\rightleftharpoons$ Circularly Polarised light

15. Define Resolving power of a grating.(May 2017)

The resolving power of a diffraction grating is defined as the capacity to form separate diffraction maxima of two wavelengths which are very close to each other. It is given by

$$\lambda/d\lambda = n.N$$

16. Define Dispersive power of a grating? (May 2018)

The dispersive power of grating is defined as the ratio of variation of angle of diffraction with wavelength. It is given by

$$d\theta/d\lambda = n / (e+d) \cos \theta$$

17. What are the advantages of Air Wedge experiment? (May 2018)

Air Wedge is used to

- i. Find the thickness of very thin materials like hair, paper
- ii. To check the flatness of a given surface

18. What is the cause of double refraction? (Dec 2019)

According to his Huygen's theory when any wavefronts strikes a doubly refracting crystal, every point of the crystal becomes a source of two wavefronts.

- a) Ordinary wavefront corresponding to ordinary rays. Since ordinary rays have same velocity in all directions, the secondary wavefront is spherical.
- b) Extra - ordinary wavefront corresponding to extra - ordinary rays. Since extra - ordinary rays have different velocities in different directions, the extra ordinary wavefront is ellipsoid of revolution, with optic axis as the axis of revolution.

19. What type of fringes will be formed if white light is used in Michelson's interferometer?

With white light, the fringes are observed only when the path difference is very small. The fringe of zero thickness is dark (i.e) central fringe and the other fringes are coloured due to overlapping of various colours.

20. Mention the uses and drawbacks of half wave plate (May 2018, Sep 2020)

It is capable of rotating the plane polarized light through 90° hence it is used as

Polarization rotator

The drawbacks of Half wave plate is it will convert elliptically polarized light into circularly polarized light and vice -versa

PROBLEMS:

1. In the Michelson Interferometer arrangement, if one of the mirrors is moved by 0.03mm, 100 fringes cross the field of view. Calculate the wavelength of light used (May 2018)

Given data:

$$d = 0.04\text{mm} = 0.03 \times 10^{-3}\text{m}$$

$$n = 100$$

Solution:

$$\begin{aligned}\text{Formula: Wavelength } \lambda &= 2d/n \\ &= \frac{2 \times 0.03 \times 10^{-3}}{100} \\ &= 6 \times 10^{-7}\text{m} \\ &= 6000\text{\AA}\end{aligned}$$

$$\text{Wavelength of light used} = 6000\text{\AA}$$

2. Calculate the least thickness of (i) A Quarter wave plate and (ii) Half wave plate for a wavelength of 6000Å°. Given $\mu_e = 1.553$ and $\mu_o = 1.544$. (May 2017)

Solution: since $\mu_e > \mu_o$ the crystal is a positive uniaxial crystal

Formulae:

(i) Minimum thickness of quarter wave plate

$$t = \lambda / 4 (\mu_e - \mu_o)$$

$$t = 6000 \times 10^{-10} / 4(1.553 - 1.544) = 1.667 \times 10^{-5}\text{m}$$

(ii) Minimum thickness of Half wave plate

$$t = \lambda / 2 (\mu_e - \mu_o)$$

$$t = 6000 \times 10^{-10} / 2(1.553 - 1.544) = 3.33 \times 10^{-5}\text{m}$$

3. An airwedge experiment is used to find the thickness of a thinsheet of paper. A monochromatic light of wavelength 5893Å is falling normally on the air wedge of length 4.6X10⁻²m. If the thickness of 5 fringes os measured to be 4X10⁻³m. Calculate the thickness of the paper (Jan 2016)

Given data:

$$\lambda = 5893\text{\AA} = 5893 \times 10^{-10}\text{m}$$

$$d = 4.6 \times 10^{-2}\text{m}$$

$$5\text{fringes} = 4 \times 10^{-3}\text{m}$$

$$\beta = 0.8 \times 10^{-3}\text{m}$$

Solution:

$$\begin{aligned}\text{Formula: } t &= \lambda / 2\beta \\ &= \frac{5893 \times 10^{-10} \times 4.6 \times 10^{-2}}{2 \times 0.8 \times 10^{-3}} \\ &= 1.694 \times 10^{-5}\text{m}\end{aligned}$$

4. A diffraction grating has 0.15m of surface ruled 6×10^5 lines/meter. What is its resolving power in the first order? (May 2019)

The resolving power of grating is $\frac{\lambda}{d\lambda} = nN$

Where N is Number of lines in the grating

n is order of diffraction

$$\frac{\lambda}{d\lambda} = 1 \times 6 \times 10^5$$

$$\frac{\lambda}{d\lambda} = 6 \times 10^5 \text{ lines / m}$$

5. In a Michelson's interferometer 200 fringes cross the field of view when the movable mirror is displaced through 0.0589mm. Calculate the wavelength of monochromatic light used.(Jan 2011)

Given data:

$$d = 0.0589 \times 10^{-3} \text{m}$$

$$n = 200$$

Solution:

Formula: Wavelength $\lambda = 2d/n$

$$= \frac{2 \times 0.0589 \times 10^{-3}}{200}$$

$$= 5.89 \times 10^{-7} \text{m}$$

$$= 5890 \text{ \AA}$$

Wavelength of light used = 5890 Å

Pondicherry University PART B Questions

- 1. Describe the construction and working of Michelson's interferometer. How this can be used for measuring the wavelength of monochromatic light? (Jan 2010, Jan 2011, Jan 2013, Jan 2014, 2016, April/May 2011, May 2012, May 2017, May 2018, May 2019, Dec 2019, Sep 2020)**

Principle :

The amplitude of light beam from a source is divided into two parts of equal intensities by partial reflection and transmission. These beams are then sent in two directions at right angles and are brought together after they suffer reflection from plane mirrors to produce interference fringes.

Construction:

It consists of two highly polished mirrors M_1 and M_2 which are at right angles to each other. There are two optically flat glass plates G_1 and G_2 of same thickness and made up of same material placed parallel to each other. These plates are inclined at an angle of 45° with the mirror M_1 and M_2 . The plate G_1 is half silvered at the back so that the incident beam divided into two beams viz, reflected beam and transmitted beam of equal intensity. The mirrors M_1 and M_2 are provided with screws on their back, so that they can be adjusted exactly perpendicular to each other.

The two rays which enter the telescope are originally derived from the same single beam; hence they cause the interference fringes in the field of view of the telescope. Hence a path difference can be introduced between the two reflected rays by moving the mirror M_1 . It is clear from the Fig. 2.7 that ray PC passes twice through the glass plate G_1 . i.e. 1st through PQ and 2nd through QP, after reflection from the mirror M_1 , whereas the ray PB does not even passes once through G_1 , even after reflection from the mirror M_2 . Thus in the absence of the glass plate G_2 the path traced by the beam between G_1 M_1 and G_1 M_2 are not equal. To equalize the path difference, a glass plate G_2 of same thickness and materials as that of G_1 is introduced between G_1 and M_2 . So that the ray PB will also pass twice, i.e. 1st through RO in glass plate G_2 and 2nd through OR in Glass plate G_2 after reflection from the mirror M_2 . The mirror M_1 is mounted on a carriage which can be moved forward and backward using the handle 'H'. The distance at which the M_1 is moved can be read with the help of the scale as shown in Fig. 2.7. The interference fringes can be observed in the field of view of the Telescope (T)

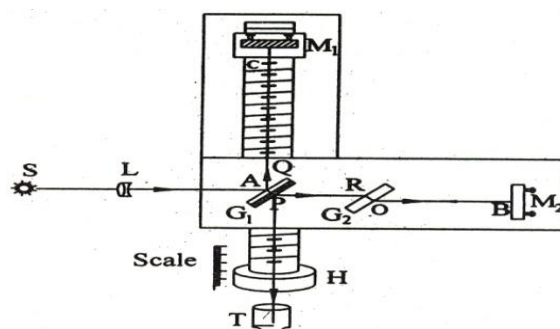


Fig. 2.7

WORKING :

Light from a monochromatic source S is made parallel with the help of collimating lens L . The light beam is allowed to fall on the semi silvered glass plate G_1 . It is partly reflected at the back surface of G_1 and travel towards M_1 i.e. along (AC) and partly transmitted towards M_2 (i.e along AB). These two rays travel along two mutually perpendicular paths and are reflected back by the mirror M_1 and M_2 . These two rays again meet at glass plate G_1 and enter a short focus telescope T . Since the glass plate G_2 is used to compensate the path difference between the two rays, it is called as a **compensation plate**. Thus the path of the two rays viz. PB and PC are made equal.

TYPES OF FRINGES

(i) Circular fringes

When mirror M_2 is exactly perpendicular to mirror M_1 and the virtual mirror M_2' (which is image of M_2) an air film of constant thickness is enclosed between them as shown in Fig. 2.8.

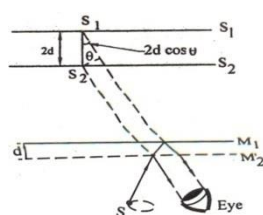


Fig. 2.8.

The air film gives reflected beam to interfere. The path difference depends upon the following factors.

- (i) The separation between M_1 and M_2'
- (ii) The angle suspended on the eye
- (ii) The inclination between M_1 and M_2'

It can be seen from the above Fig.2.8 that the path differences are different for different values of θ . The source S is an extended source and S_1 and S_2 are the virtual source due to M_1 and M_2' . If the distance between M_1 and M_2' is d and the distance between S_1 and S_2 is $2d$, then

The path difference between the two rays = $2d \cos \theta$

The equation for maxima is given by $2d \cos \theta + \frac{\lambda}{2} = n\lambda$, Where $\frac{\lambda}{2}$ is the additional path difference between the rays

Case (i)

When M_1 and M_2' coincide, the path difference is zero and the field of view is perfectly dark as shown in Fig. 2.9

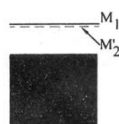


Fig. 2.9

Case (ii)

When M_1 is moved either way parallel to itself, widely spaced circular fringes are produced. The width of the fringes depends upon the path difference between the rays Fig. 2.10.

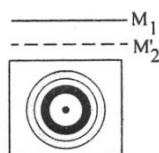


Fig. 2.10

(i) Localised fringes

When the mirror M_2 is not exactly perpendicular to M_1 or if M_1 and the vertical Mirror M_2' are inclined then the air film enclosed between them is wedge shaped. For small path difference, the fringes are nearly straight but for large path difference the fringes are generally curved and are always convex towards the thin edge of the wedge. When the two mirrors M_1 and M_2' intersect in the middle, straight line fringes are observed as shown in Fig. 2.11(a). When the two mirrors are inclined, curved fringes are observed as shown in Fig. 2.11 (b&c)

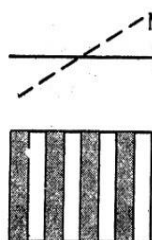


Fig. 2.11 (a)

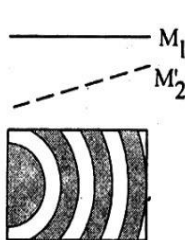


Fig. 2.11 (b)

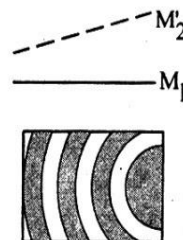


Fig. 2.11 (c)

(ii) Localised white light fringes

With white light, the fringes are observed only when the path difference is very small. The fringe of zero thickness is dark (i.e) central fringe and the other fringes are coloured due to overlapping of various colours. If the film is thick, uniform illumination is observed. White light fringes are useful to

DETERMINATION OF WAVE LENGTH OF A MONOCHROMATIC SOURCE OF LIGHT

The mirrors M_1 and M_2 are adjusted to get the circular fringes in the field of view of the telescope. Then the mirror M_1 is moved with the help of handle of the micrometer screw and the number of fringes that cross the field of view is counted. If 'n' is the number of fringes that cross the field and 'd' is the distance through the mirror is moved, then, from the theory of circular fringes.

$$2d \cos\theta = n\lambda \text{ (Bright)}$$

Since θ is very small and is ≈ 0

$$\therefore 2d = n\lambda$$

$$d = \frac{n\lambda}{2}$$

$$\text{or } \lambda = \frac{2d}{n}$$

If $n = 1$ (i.e) only one fringe crosses the centre of field of view,
then $d = \frac{\lambda}{2}$ (i.e) the mirror moves through a distance which is equal to half of the wavelength
 $\therefore \lambda = 2d$

2. Deduce an expression for air-wedge to find the thickness of a thin sheet of paper or wire. (Jan 14)

Deduce an expression for Air Wedge to find the thickness of a thin sheet of paper, and hence explain an experimental method to find the thickness of a thin sheet of a paper using Air Wedge Experiment? (April 2010, May 2011, May 2019)

Principle

Two plane glass plates are inclined at an angle (θ) by introducing a thin material (e.g. hair), forming a wedge shaped air film. The film is illuminated by sodium light. Interference occurs between the two rays, one reflected from the front surface and the other by internal reflection at the back surface. Therefore straight line fringes parallel to the edge of the wedge are obtained. Using the theory of air wedge the thickness of the material can be determined.

DESCRIPTION

Two optically plane glass plates are placed one over the other and tied by means of a rubber band at one end. The given material of wire or paper is introduced on the other end, so that an air wedge is formed between the plates as shown in Fig. 2.3.

Experimental set up:

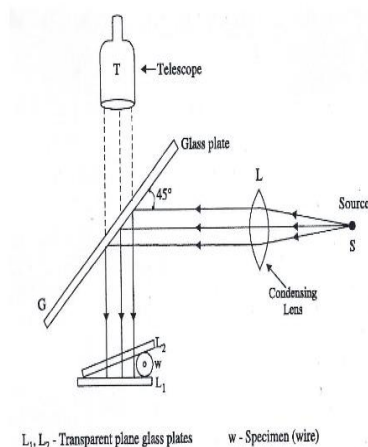


Fig. 2.3

This set up is placed on the horizontal bed plate of the travelling microscope. Light from the source 'S' can be made to fall on the Air Wedge set up with the help of a condensing lens (L) and 45° angled glass plate (G)

WORKING

Light from the sodium vapour lamp (S) is rendered parallel by means of a condensing lens (L). The parallel beam of light is incident on a plane glass plate (G) inclined at an angle of 45° and gets reflected. The reflected light is incident normally on the glass plates in contact.

Interference takes place between the light reflected from the top and bottom surfaces of the glass plates and is viewed through the travelling microscope (M). Since the thickness of the materials, remains constant in the direction parallel to the thin edge of the wedge, a large number of equally spaced dark and bright straight line fringes are formed parallel to the edge of contact as shown in Fig. 2.4.

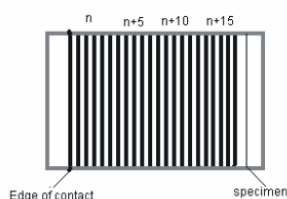


Fig. 2.4

The microscope is adjusted so that the bright (or) dark fringe near the edge of contact is made to coincide with the vertical cross wire and this is taken as the n^{th} fringe. The reading from the horizontal scale of the travelling microscope is noted. The microscope is moved across the fringes using the horizontal transverse screw and the readings are taken when the vertical cross wire coincides with every successive 5 fringes (5, 10, 15, 20.....) The width of every 20 fringes is calculated and the width of one fringe is calculated. The mean of this gives the fringe width (β)

We know from the theory of Air wedge, the fringe width $\beta = \frac{\lambda}{2\theta}$ (1)

If 'l' is the distance between the edge of contact and the material (say the distance of the last fringe) and 't' is the thickness of the material, then
For θ being very small from the Fig. 2.5

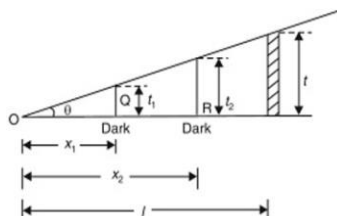


Fig. 2.5

From Fig. 2.5 We can write $\theta = \frac{t}{l}$ (2)

Substituting eq (2) in (1) we get

$$\text{Fringe width } \beta = \frac{\lambda l}{2t}$$

$$\therefore \text{Thickness of the material } t = \frac{\lambda l}{2\beta}$$

Thus, by finding λ , l and β the thickness of any thin material can be determined using air wedge.

3. Explain briefly the resolving power of grating and prism (Jan 2014, April/May 2011, May 2019)

RESOLVING POWER OF A PRISM

The term resolving power is applied to spectrographic devices using a prism or a grating. Resolving power signifies the ability of the instrument to form separate spectral images of two neighboring wavelengths λ and $\lambda + d\lambda$ in the wavelength region λ . In the Fig.2.18, S is a source, of light, L_1 is a collimating lens and L_2 is the telescope objective.

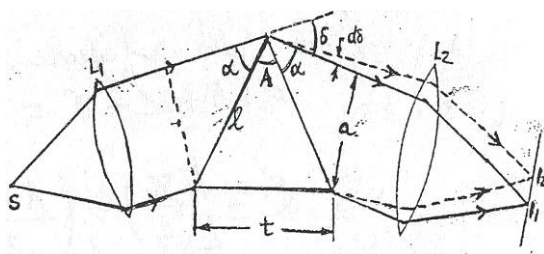


Fig.2.18

As the two wavelength λ and $\lambda + d\lambda$ are very close, if the prism is set in the minimum deviation position it would holds good for both the wavelengths the final image I_1 corresponds to the principal maximum for wavelength λ and image I_2 corresponds to the principal maximum for wavelength $\lambda + d\lambda$. I_1 and I_2 are formed at

the focal plane of the telescope objective lens L_2 . The face of the prism limits the incident beam to a rectangular section of width 'a'. Hence, the Rayleigh criterion can be applied in the case of a rectangular aperture.

In the case of diffraction at a rectangular aperture, the position of I_2 will correspond to the first minimum of the image I_1 for wavelength λ provided. a. $d\delta = \lambda$

$$d\delta = \frac{\lambda}{a} \dots\dots\dots(1)$$

From the Figure.2.18.

$$\alpha + A + \alpha + \delta = \pi$$

$$\alpha = \left[\left(\frac{\pi}{2} \right) \right] - \left(\frac{A + \delta}{2} \right)$$

$$\sin \alpha = \sin \left[\frac{\pi}{2} - \left(\frac{A + \delta}{2} \right) \right]$$

$$\sin \alpha = \cos \left(\frac{A + \delta}{2} \right)$$

$$\text{But } \sin \alpha = \frac{a}{l}$$

$$\therefore \cos \left(\frac{A + \delta}{2} \right) = \frac{a}{l} \dots\dots\dots (2)$$

$$\text{Also } \sin \frac{A}{2} = \frac{t}{2l} \dots\dots\dots (3)$$

In the case of a prism

$$\mu = \frac{\sin \left(\frac{A + \delta}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

$$\sin \left(\frac{A + \delta}{2} \right) = \mu \sin \frac{A}{2} \dots\dots\dots (4)$$

Here μ and δ are dependent on wavelength of light λ

Differentiating equation (4) with respect to λ

$$\frac{1}{2} \left(\cos \left[\frac{A + \delta}{2} \right] \right) \frac{d\delta}{d\lambda} = \frac{d\mu}{d\lambda} \left[\sin \left(\frac{A}{2} \right) \right] \dots\dots\dots (5)$$

Substituting the values from equations (2) and (3) in eq (5)

$$\frac{1}{2} \left(\frac{a}{l} \right) \frac{d\delta}{d\lambda} = \frac{d\mu}{d\lambda} \left(\frac{t}{2l} \right)$$

$$a. \frac{d\delta}{d\lambda} = t. \frac{d\mu}{d\lambda} \dots\dots\dots (6)$$

Substituting the values of $d\delta$ from equation (1)

$$\frac{\lambda}{d\lambda} = t. \frac{d\mu}{d\lambda} \dots\dots\dots (7)$$

The expression $\frac{\lambda}{d\lambda}$ measures the resolving power of the prism. It is defined as the ratio of the wavelength of a spectral line to the difference in wavelength between this line and a neighboring line such that these lines are just resolved according to **Rayleigh's Criterion**.

$\therefore$ Resolving power of a prism is given by $t \frac{d\mu}{d\lambda}$

Thus, the resolving power of a prism is directly proportional to

- (i) Length of the base of the prism and
- (ii) Rate of change of refractive index with respect to wavelength for that particular material.

The expression for resolving power given above is applicable only to spectral lines of equal intensity. If two spectral lines are of different intensities, then the value of $d\lambda$ i.e., the difference in wavelength between the two lines must be higher so that the two lines appear as separate ones.

RESOLVING POWER OF A PLANE DIFFRACTION GRATING:

The resolving power of a grating is defined as the ratio of the wavelength of any spectral line to the difference in wavelength between this line and a neighboring line such that the two lines appear to be just resolved. Thus the resolving power of a grating is equal to $\frac{\lambda}{d\lambda}$

In the Fig. 2.19, XY is the grating surface and MN is the field of view of the telescope. P_1 is the n^{th} primary maximum of a spectral line of wavelength λ at an angle of diffraction θ_n , P_2 is the n^{th} primary maximum of a spectral line of wavelength $\lambda + d\lambda$ at an angle of diffraction $\theta_n + d\theta$

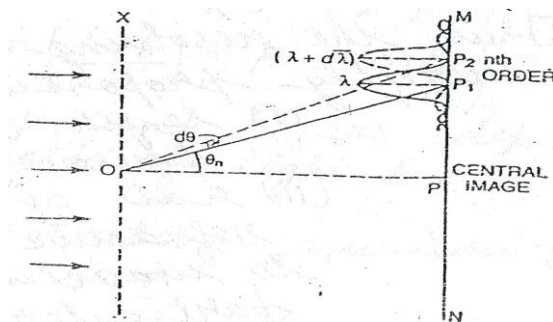


Fig. 2.19

Let P_1 and P_2 are the spectral lines in the n^{th} order. These two spectral lines according to Rayleigh, will appear resolved if the position of P_2 also corresponds to the first minimum of P_1 .

The direction of the n^{th} primary maximum for a wavelength λ is given by

$$(a+b) \sin \theta_n = n\lambda \dots\dots\dots(1)$$

The direction of the n^{th} primary maximum for a wavelength $(\lambda + d\lambda)$ is given by
 $(a+b) \sin (\theta_n + d\theta) = n(\lambda + d\lambda)$ (2)

The two lines will appear just resolved if the angle of diffraction $(\theta_n + d\theta)$ also corresponds to the direction of the first secondary minimum after the n^{th} primary maximum at P_1 (corresponds to wavelength λ). This is possible if the extra path difference introduced is $\frac{\lambda}{N}$ where N is the total number of lines on the grating surface.

$$\therefore (a+b) \sin (\theta_n + d\theta) = n\lambda + \frac{\lambda}{N}$$
(3)

Equating, the RHS of equation (2) and

$$\begin{aligned} n(\lambda + d\lambda) &= n\lambda + \frac{\lambda}{N} \\ nd\lambda &= \frac{\lambda}{N} \\ nd\lambda &= \frac{\lambda}{N} \end{aligned}$$
(4)

The quantity $\frac{\lambda}{d\lambda} = nN$ measures the resolving power of a grating. Thus, the resolving power is independent of the grating constant. The resolving power is directly proportional to

- (i) the order of the spectrum and
- (ii) the total number of lines on the grating surface.

For a given grating, the spacing between the spectral lines is double in the second order than that in the first order.

4. (a) Explain Quarter and Half wave plates in detail. (Jan 2014, May 2018, Dec 2019)

QUARTER WAVE PLATE :

Quarter wave plate is doubly refracting material made of uniaxial crystal whose refracting faces are cut parallel to the optic axis. The thickness of this plate is in such a way that it produces a path difference of $\frac{\lambda}{4}$ (or) a phase difference of $\frac{\pi}{2}$ between the ordinary and extra ordinary rays, emerging from it.

Thickness of Positive uniaxial crystal

In the case of positive uniaxial crystal of thickness 't' the velocity of O-ray is greater than the velocity of E-ray. Therefore, the refractive index of O-ray will be lesser than the E-ray.

$$\begin{aligned} \therefore \mu_e &> \mu_o \\ \therefore \text{Path difference} &= t(\mu_e - \mu_o) = \frac{\lambda}{4} \end{aligned}$$

$$(or) \text{ Thickness } t = \frac{\lambda}{4(\mu_e - \mu_o)}$$

Thickness of negative uniaxial crystal

In the case of negative uniaxial crystal of thickness 't', the velocity of E-ray is greater than O-ray. Therefore, the refractive index of E-ray, is lesser than the O-ray ($\mu_o > \mu_e$).

$$\therefore \text{ Path difference } = t(\mu_o - \mu_e) = \frac{\lambda}{4}$$

$$(or) \text{ Thickness } t = \frac{\lambda}{4(\mu_o - \mu_e)}$$

Applications :

(i) When the vibrations of the polarised light, inclined at an angle of 45° to the optics is made to incident on a quarter wave plates the emergent beam becomes circularly polarised light. Therefore quarter wave plate is used to convert plane polarised light to circularly polarised light and vice versa.

(ii) When the vibrations of the polarised light inclined at an angle not equal to 45° (i.e. other than 45°) to the optic axis is made to incident on a quarter wave plate, the emergent beam becomes elliptically polarised light. Thus it is used to convert plane polarised light to elliptically polarised light and vice versa.

Thus a quarter wave plate can be used to convert.

(i) Plane polarised light $\rightleftharpoons$ Elliptically polarised light

(i) Plane polarised light $\rightleftharpoons$ Circularly Polarised light

Half Wave Plates

Definitions :

Half wave plate is a doubly refracting material made of uniaxial crystal like quartz (or) calcite, whose refracting faces are cut parallel to the optic axis. The thickness of this plate is in such a way that it produces a path different of $\frac{\lambda}{2}$ or phase difference π between the ordinary - ray and the extra ordinary ray emerging from it

$$\text{Thickness of positive uniaxial crystal is } t = \frac{\lambda}{2(\mu_e - \mu_o)}$$

$$\text{Thickness of negative uniaxial crystal is } t = \frac{\lambda}{2(\mu_o - \mu_e)}$$

APPLICATION :

When a plane - polarised light is incident on a half - wave plate such that it makes an angle of 45° with the optic axis, then the emergent beam becomes a plane - polarised light with 90° rotation. Thus the half wave plate is capable of rotating the angle of the plane polarised light by 90°

(b) Write a short note on Dispersive power of grating.

DISPERSIVE POWER OF A GRATING

Dispersive power of a grating is defined as the ratio of the difference in the angle of diffraction of any two neighboring spectral lines to the difference in wavelength between the two spectral lines. It can also be defined as the difference in the angle of diffraction per unit change in wavelength. The diffraction of the n^{th} order principal maximum for a wavelength λ is given by the equation,

$$(a + b) \sin \theta = n\lambda \text{ -----(1)}$$

Differentiating this equation with respect to θ and λ ($a + b$) is constant and n is constant in a given order

$$(a + b) \cos \theta d\theta = n d\lambda$$

$$\frac{d\theta}{d\lambda} = \frac{n}{(a + b) \cos \theta}$$

$$\frac{d\theta}{d\lambda} = \frac{nN'}{\cos \theta} \text{ ----- (2)}$$

In the equation (2) $\frac{d\theta}{d\lambda}$ is the dispersive power, n is the order of the spectrum,

N' is the number of lines per cm of the grating surface and θ is the angle of diffraction for the n^{th} order principal maximum of wavelength λ . From equation (2), it is clear that the dispersive power of the grating is

- (i) Directly proportional to the order of the spectrum
- (ii) Directly proportional to the number of lines per cm and
- (iii) Inversely proportional to $\cos \theta$.

5. Briefly explain the following: (Jan 2013, 2016, May 2019)

(a) Antireflection coating

A transparent material whose refractive index value lies between air and glass is coated on the surface the glass in order to prevent or to reduce reflection is called as anti reflection coating.

(b) Optical activity.

The phenomenon or the property of rotating the plane of vibration by certain crystals or substances is known as optical activity.

(c) Explain double refraction using Huygen's theory (May 2017, May 2018, May 2019, Sep 2020)

HUYGEN'S THEORY OF DOUBLE REFRACTION

According to his Huygen's theory when any wavefronts strikes a doubly refracting crystal, every point of the crystal becomes a source of two wavefronts.

- a) Ordinary wavefront corresponding to ordinary rays. Since ordinary rays

have same velocity in all directions, the secondary wavefront is spherical.

b) Extra - ordinary wavefront corresponding to extra - ordinary rays. Since extra - ordinary rays have different velocities in different directions, the extra ordinary wavefront is ellipsoid of revolution, with optic axis as the axis of revolution.

The sphere and ellipsoid touch each other at points which lie on the optic axis of the crystal, because the velocity of ordinary and extra - ordinary ray is same along the optic axis. In certain crystals (like calcite and tourmaline) called the **negative crystals**, the ellipsoid lies outside the sphere as shown in Fig. 2.21 (a)

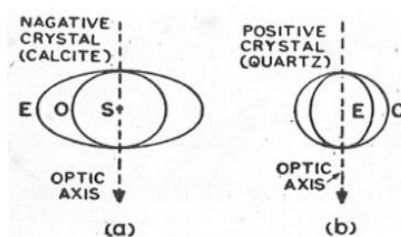


Fig. 2.21 (a) & b

This shows that in negative crystals, the extra ordinary wavefront travels faster than ordinary wavefront except along optic axis. In certain crystals (likethose of ice and quartz) called the positive crystals, the sphere lies outside the ellipsoid as shown in Fig. 2.21 (b). This shows that velocity of ordinary wavefront is greater than extra - ordinary wavefront (in positive crystals) except along optic axis.

6. Explain the term diffraction grating? (4) (May 2013)

Diffraction grating or plane transmission grating is an arrangement consisting of a large number of parallel and equidistant slits of the same width (10^{-6}m) on a transparent material such as glass, with a diamond pointer. The ruled lines are opaque to light while the space between any two lines is transparent to light and act as grating. This is known as plane transmission grating.

7. With neat diagram explain the construction and working of Laurent's half shade Polarimeter and explain the method to determine the specific rotation of sugar solution. (May 2017, Dec 2018, May 2019, Dec 2019)

Polarimeters are the instruments designed to measure the angle of rotation produced by a substance. When they are calibrated to read directly the percentage of cane sugar in a solution, they are named as Saccharimeters. Polarimeters can be used to find the specific rotation of sugar solution or if the specific rotation is known, they can be used to find its concentration.

CONSTRUCTION

The essential parts of a Laurent's half - shade Polarimeter as shown in Fig. 2.32

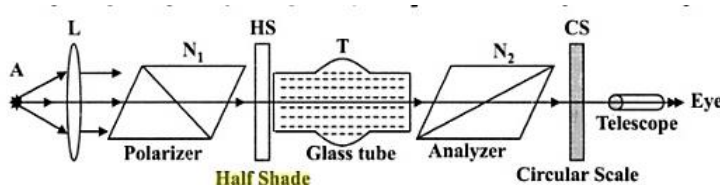


Fig. 2.32

'S' is a monochromatic source of light and L is a convex lens which renders the incident light into a parallel beam. N_1 and N_2 are two nicol prisms. N_1 acts as polariser while N_2 act as analyser. N_2 is capable of rotation about a common axis of N_1 and N_2 . The rotation of analyser (N_2) can be read on a graduated circular scale (CS.). The verniers are also provided to read the fraction of a degree. Light after passing through polariser becomes plane polarised with its vibrations in the principal plane of the Nicol N_1 . The plane polarised light now passes through a half shade device (H.S.) and then through a tube T containing the optically active substance (say sugar solution.)

Usually T is a hollow glass tube having a large diameter in the middle so that no air bubble may be in the path of light when filled with a liquid. This tube is closed at the ends by cover - slips and metal covers. The emergent light on passing through analyser N_2 is viewed through a telescope T. The telescope is focussed on the half - shade.

ACTION OF HALF - SHADE

When an optical active substance is placed in between two crossed Nicols the field of view is not dark. In order to make the field of view dark, the analyser is rotated. It is observed that when the analyser is rotated the field of view is not dark, for a considerable region. Hence the measurement of optical rotation is not accurate. To avoid this difficulty a half shade device is used. The Laurent's half - shade plate consists of a semi circular half - wave plate ABC of quartz (cut parallel to optic axis) so that it introduces a phase change of π between extra - ordinary and ordinary rays passing through it and a semicircular glass plate ADC as shown in Fig. 2.33.

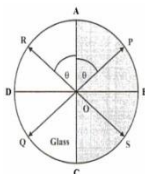


Fig. 2.33.

The two plates are cemented along AC. The thickness of glass plate is such that it absorbs the same amount of light as the quartz plate. Let the plane of vibration of the plane polarized light incident normally on the half shade plate is along PQ. PQ makes an angle θ with AC as in Fig. 2.33. The vibrations emerge from glass plate along PQ. Inside quartz plate light is divided into two components, one, O-ray along XX', and the other, E-ray, parallel to optic axis, YY' as shown in Fig.2.34. The two components travel along same direction but with different speeds. As quartz is a positive uniaxial doubly refracting crystal.

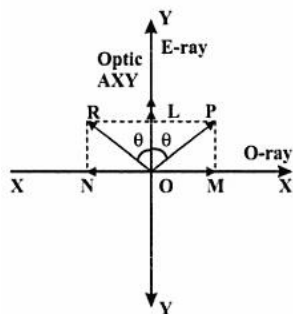


Fig. 2.34

E-ray component moves with higher velocity than O-ray component and on emergence a phase difference of π introduced between them. Due to this phase difference the direction of O components reversed. If initial direction is OM, then, it finally becomes ON. So, the resultant of E component OL and ON will be OR which makes an angle θ with Y-axis.

The vibration of beam of emergent light out of quartz will be along RS. If the principal plane of N_2 is along PQ, the light pass unobstructed through glass plate ADC, but partly obstructed by ABC, so the glass half will appear brighter than the quartz half as shown in Fig. 2.35 (a). Other way, if the principal plane of N_2 is along RS, reverse situation occurs, so then quartz half will appear brighter than the glass half as shown in Fig. 2.35(b).

However, if the principal plane of N_2 is parallel to AC (Y-axis), it is equally inclined both for O and E rays. The field of view becomes equally illuminated as shown in Fig. 2.35(c). Thus, half shade serves the purpose of dividing the field of view in two halves. When there is a rotation of N_2 from the position of equal brightness or illumination a marked change in the intensity of two halves is observed.

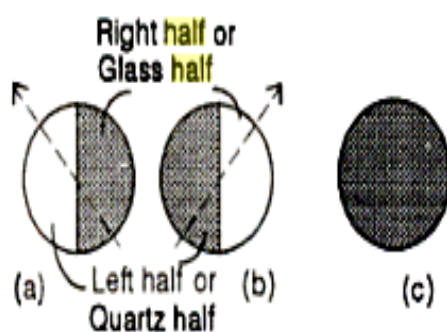


Fig. 2.35 (a) 2.35 (b) 2.35 (c)

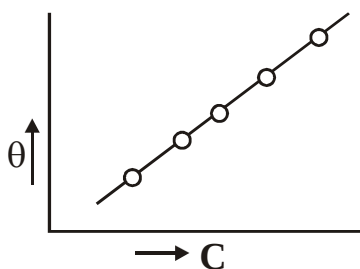
Determination of specific Rotation of sugar solution by Laurents Half shade polarimeter

We know that the specific rotation S is given by

$$S = \frac{10\theta}{LC} \text{ deg/dm cc}$$

Where θ is the angle of rotation in degrees, l is the length of the solution in decimeters and C is the concentration of solution in gm per cc. Hence to determine the specific rotation of a substance solution of known concentration is prepared. The length of the solution is measured directly. The value of θ is determined as follows. First of all the experimental tube is filled with water and placed in its position. The telescope is focused on the half shade and the analyser is rotated till equally bright position is obtained. The readings of two verniers are noted on the circular scale. Now the tube is filled with the optically active solution already prepared and placed it in its position. The field of view in this position is equally bright. The analyser is rotated and is brought to a position so that the whole field of view is equally bright. The new positions of the two verniers are again noted on the circular scale. The difference in the two readings of the same verniers gives the angle of rotation produced by the solution. The experiment is repeated with different concentrations of the solution.

A graph is then plotted between concentration and angle of rotation as shown below. This comes out a straight line.



From the graph the ratio θ/C is determined. Substituting these values, the specific rotation is calculated by using the above formula.

8. What is meant by dispersive power? Derive a mathematical relation for the dispersive power of grating. (Dec 2018, May 2018)

DISPERSIVE POWER OF A GRATING

Dispersive power of a grating is defined as the ratio of the difference in the angle of diffraction of any two neighboring spectral lines to the difference in wavelength between the two spectral lines. It can also be defined as the difference in the angle of diffraction per unit change in wavelength. The diffraction of the n^{th} order principal maximum for a wavelength λ is given by the equation,

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Differentiating this equation with respect to θ and λ ($a + b$) is constant and n is constant in a given order

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- (iv) Directly proportional to the order of the spectrum
- (v) Directly proportional to the number of lines per cm and
- (vi) Inversely proportional to $\cos \theta$.